Automating the “napkin sketch” in optical design

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ABSTRACT

Optical design typically begins with a manually chosen architecture comprising optical components and their geometric arrangement. While refinement of a prescribed architecture has largely been automated, discovering the architecture itself remains difficult to formulate as a general computational task.

Given an optical task and user-defined specifications, we initialize a design environment and run an adaptive search mechanism over the light path geometries and interaction sequences. We explore candidate architectures without prescribing a fixed light path or sequence of optical interactions. This formulation allows us to search over a richer space of solutions with diverse types and sequences of interactions. Chosen candidates are refined using a tolerance-aware optimization process.

We demonstrate the approach across multiple optical-design problems via novel optical solutions. These resulting designs include a spectrometer with [X percent] at [spectral resolution], a night-vision system achieving [MTF at spatial frequency] with [maximum distortion] across [field of view], and a thermal imager with [RMS spot radius] over [field and wavelength range]. Under [specified fabrication and alignment tolerances], [design/system] retains [performance criterion] in [Y percent of trials]. Our framework achieves wide adaptability to user specifications and is competent in meeting performance requirements.

We also highlight the significant time advantage offered by our automated discovery framework. On average, our systems required [time] to proceed from user-specification entry to tolerance-ready designs. We additionally provide the designs and corresponding performance characterizations for a selected few designs.

Introduction

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For the last four hundred years, optical design has begun with the creative conception of a starting point [1, 2]. This involves designers discovering a satisfactory combination of the light path, necessary sequences of optical interactions, and the arrangement of optical components. In practice, designers often instead consult existing successful architectures developed for familiar applications. Downstream refinement, which makes use of advances in numerical optimization [3] and deep-learning-enabled methods [4], takes these starting points toward achieving the desired optical performance. However, for novel applications requiring ingenuity in finding new ways to route light, identifying a suitable starting configuration still requires significant creativity. The choice of starting configuration, even for well-explored architectures, is particularly important, as it can substantially affect the achievable optical performance even after extensive numerical optimization [5].

Conventional optical designs often deal with a fixed sequence of interactions, belonging to a particular baseline optical architecture upon which newer iterations are developed and refined. However, unlocking novel capabilities often requires discovering new sequences of optical interactions within the geometric constraints imposed by an application. *Pancake optics* [6, 7], for example, make use of a sequence of polarization-dependent reflections, transmissions, and refractions to "fold" light into the compact volume required by near-eye displays. Systemizing a way to incorporate multiple interaction types and discover diverse sequences can therefore, be an effective way to discover novel optical architectures that make a generational leap in performance.

Off-axis and folded optical systems can achieve high optical performance by incorporating freeform elements. Early work explored physics-guided approaches for constructing and refining these systems, including point-by-point surface construction [8], aberration-guided starting geometry generation [5], and Fermat’s principle-based direct design [9]. At the broader design level, other work systematically explored the three-mirror solution space [10].

A parallel body of work has sought to automate freeform design through reinforcement learning [11], deep-learning-based starting-point generation [12], multiple-solution generation with FreeformNet [13], and more recently, conditional generative models [14].

While these methods have yielded successful, high-quality designs for their respective applications, they generally do not explore performance improvements through diversity in the types and sequences of optical interactions themselves. Much of the current work continues to build upon established optical architectures, seeking performance gains through reparameterization and optimization, while leaving the interaction sequences comparatively underexplored.

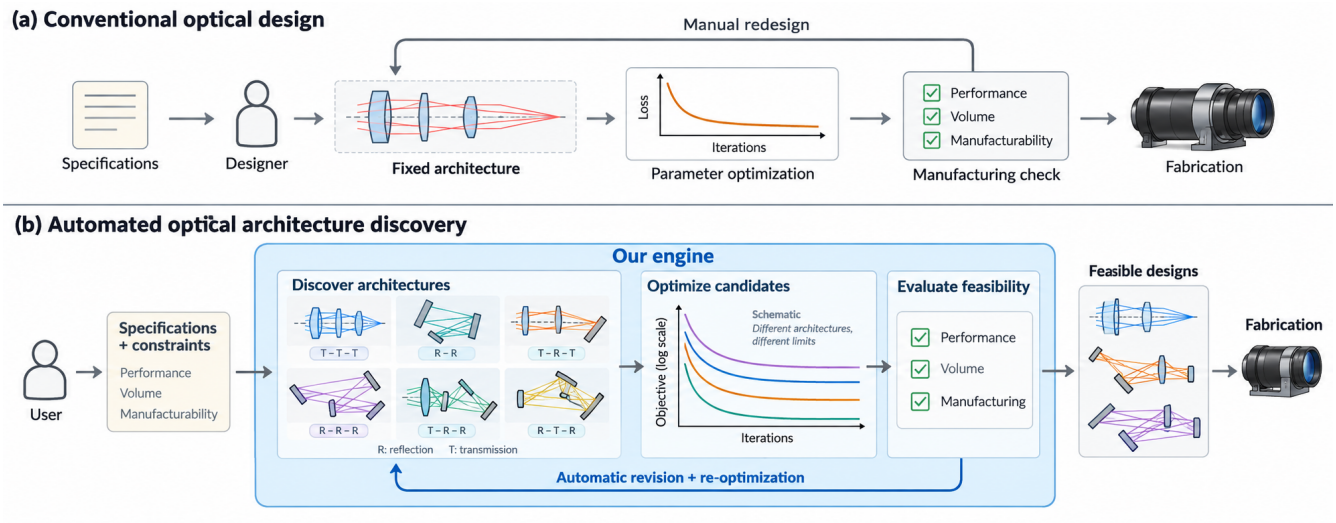


Figure 1. Overview

Results

Discussion and Conclusion

Methods

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Author contributions statement

Competing interests

The authors declare no competing interests.

Supplementary Information

Supplementary Information accompanies this manuscript as part of the submission files.